

ROBERT SICHOMBA

Geoscientist | Data Scientist | Geophysicist | Geotechnical & Materials Specialist

North Western, Zambia | +260 970312455 | sichombarobertbob@gmail.com

[in](#) LinkedIn [G](#)itHub [P](#)ortfolio

Professional Summary

Multidisciplinary Geoscientist and Data Scientist with over 3 years of combined experience in **exploration geology, geophysics, geotechnical engineering, and advanced data analytics**. Proven ability to apply **Python, SQL, machine learning, and geostatistics** to geological, geophysical, and engineering datasets. Experienced in managing the full exploration pipeline—from target generation, diamond drilling supervision, and core logging to resource estimation, ore control, and multivariate geochemical analysis. Skilled in integrating geophysical datasets (Magnetics, IP), structural geology, and GIS-based spatial modelling. Currently completing an **MSc in Data Science and Exploration Geology** (Kobold Metals sponsored). Combines rigorous field expertise with advanced data science to deliver actionable insights and drive data-driven decision-making. Seeking opportunities in **geology, data science, statistics, physics, or geotechnical engineering**.

Core Competencies

- Geology & Exploration:** Diamond Drilling, Core Logging, Soil/Stream Sediment Sampling, Geological Mapping, Structural Geology, Surveying, Resource Estimation, Ore Control, Grade Control, QA/QC.
- Geophysics:** Magnetics, Induced Polarization (IP), Magnetic Susceptibility (Mag Sus), Gamma Spectrometry, Downhole Logging, Portable XRF (pXRF), Geophysical Data Processing & 2D/3D Interpretation.
- Data Science & Statistics:** Python (pandas, numpy, scikit-learn, matplotlib, streamlit), SQL, PostgreSQL/PostGIS, Machine Learning, Geostatistics, Compositional Data Analysis (CODA), PCA, Clustering, Regression, Anomaly Detection, Statistical Process Control, Data Visualization, Dashboards.
- Geotechnical & Materials:** Site Investigation, Borehole Logging, SPT, In-situ Testing, Borrow Pit Exploration, Soils & Aggregates, Asphalt/Concrete Testing, Compaction Control, ASTM, BS, AASHTO, RDA, SANS 3001, ISO 17025, Marshall Mix Design.
- Physics & Teaching:** Laboratory Physics, Experimental Design, Data Analysis, Teaching & Mentoring (100+ students), Python for Data Analysis.
- GIS & Spatial Analysis:** QGIS, ArcGIS, Spatial Interpolation, Georeferencing, Cross-sections, 3D Visualisation.
- Management & Strategy:** Project Management, Team Leadership (10+ technicians), Stakeholder Liaison, Technical Reporting, QA/QC Protocols.

Education

Master of Science in Data Science and Exploration Geology 2024 – August 2026
The University of Zambia and The Copperbelt University (Kobold Metals Sponsored Masters Program)

Geostatistics, machine learning, database design, exploration targeting, CODA, multivariate analysis

Bachelor of Science in Geophysics (With Merit) 2017 – 2022
Copperbelt University, Kitwe, Zambia
Geology, geophysics, materials science, experimental design, data analysis

Zambia School Certificate (O Levels) 2013 – 2015
Chunga High School, Lusaka, Zambia

Professional Experience

Project Geologist — Geo-Junction Exploration, Zambia 2024 – Present

- Manage the **full exploration pipeline**: soil sampling, stream sediment surveys, **diamond drilling supervision**, meter marking, **core logging (lithological, structural, mineralization)**, and QA/QC of sample preparation.
- Conduct **magnetic susceptibility (Mag Sus)**, **portable XRF (pXRF)**, and **geophysical surveys (Magnetics, IP)** for rapid lithological and geochemical characterisation.
- Assist in **resource estimation** and **ore control** by integrating geological, geophysical, and assay data into 3D models.
- Design and manage exploration databases using **PostGIS**; implement robust data validation and QA/QC protocols (CRMs, blanks, duplicates).
- Apply **Compositional Data Analysis (CODA)** and multivariate statistics (PCA, clustering, regression) to identify geochemical anomalies and lithogeochemical signatures.
- Set up **QGIS projects end-to-end**; produce spatial maps, interpolated geochemical grids (RBF), cross-sections, and 3D visualisations for drill target generation.
- Compile weekly exploration reports and present integrated interpretations to senior management, leading to **successful drill targeting and strategic decision-making**.

Assistant Geotechnical & Materials Project Manager — AVIC International, Lusaka–Ndola Road Project 2023 – 2024

- Led a geotechnical and materials laboratory team of **10+ technicians** across multiple work fronts, ensuring timely delivery of test results and **98% compliance** with RDA/ASTM/BS specifications.
- Coordinated drilling, sampling, and field investigation schedules; supervised borehole logging, SPT, and in-situ testing.
- Implemented **ISO 17025** principles; introduced statistical process control and automated QA/QC reporting using Python – achieved **40% time savings** on reporting.
- Led borrow pit exploration and characterisation (**24+ pits**); evaluated material quality and reserves, recommended optimal sources saving **15% haulage costs**.
- Liaised with client engineers, project directors, and subcontractors; managed non-conformance reporting and corrective actions.

Undergraduate Lecturer Assistant (Physics & Data Science) — Copperbelt University, Kitwe 2023

- Taught Python for data analysis, laboratory physics, and statistics – mentored **100+ students**.

Data Science Intern — Zambia Statistics Agency, Lusaka 2022

- Analysed large census datasets using Python and SQL; developed ETL pipelines and dashboards.

Technical Expertise

- **Exploration & Mining Geology:** Diamond drilling supervision, core logging (structural, lithological, mineralisation), soil/stream sediment/rock chip sampling, geological mapping, structural measurements, surveying, resource estimation, ore control, grade control.
- **Geophysics & Interpretation:** Magnetics, Induced Polarization (IP), magnetic susceptibility (Mag Sus), portable XRF (pXRF), gamma spectrometry, downhole logging, geophysical data processing and 2D/3D interpretation.
- **Geochemical Data Science:** Compositional Data Analysis (CODA), multivariate statistics (PCA, PLS, clustering, random forest, SVM), anomaly detection, variography, kriging, machine learning for mineral prospectivity mapping.
- **GIS & Spatial Analysis:** QGIS (end-to-end project setup, map production, spatial interpolation, georeferencing, cross-sections, 3D visualisation), ArcGIS (basic), map package generation.
- **Database & Programming:** PostgreSQL/PostGIS, Python (pandas, numpy, scikit-learn, matplotlib, streamlit, pyproj), SQL, Git, SGeMS, R (basic), LaTeX.
- **Data Science & Analytics:** Machine Learning, Statistical Modelling, Data Mining, Data Visualization, Dashboard Development (Plotly, Streamlit), ETL Pipelines.
- **Geotechnical & Materials:** Site investigation, soil/rock mechanics, materials testing (ASTM, BS, AASHTO, RDA, SANS 3001), ISO 17025, Marshall Mix Design, nuclear gauge, concrete testing, compaction control.
- **Physics & Teaching:** Laboratory physics, experimental design, data analysis, teaching, mentoring.

Key Achievements

- Successfully identified and prioritised **5 high-confidence drill targets** using RBF interpolation and PCA on a 2,000-soil-sample geochemical dataset.
- Integrated **Magnetics and IP geophysical data** with geological mapping to refine structural interpretations and optimize drill hole placement.
- Assisted in **resource estimation and ore control** initiatives, ensuring accurate grade boundary delineation and minimizing dilution.
- Automated QA/QC reporting using Python, reducing reporting time by **40%** and achieving **98% specification compliance**.
- Discovered and characterised **24+ borrow pits**, recommending 8 sources that saved **15% haulage costs**.
- Designed and implemented a PostgreSQL exploration database tracking over **500+ samples** with automated validation and reporting.
- Built interactive spatial mapping dashboards (Plotly) for real-time drill targeting and exploration review.

Certifications & Training

- ISO 17025:2017 Laboratory Quality Management – 2025
- Marshall Mix Design (ASTM D6926/D6927) – 2025
- SANS 3001 – Soils and Aggregates Testing – 2025
- Nuclear Gauge Safety (Troxler) – 2025
- Concrete Testing (JTG 3420-2020) – 2025
- Advanced Hydrogeology – Fulbright Program – 2025

References

- **Felix Tan** — Chief Laboratory Engineer, AVIC International Zambia Ltd
Phone: +260 979574375
- **Mr. Mbasela** — Director, Geo-Junction Exploration
Phone: +260 973871998
- **Mr. Amon George Sikombe** — Supervisor, Zambia Statistics Agency
Phone: +260 964855869

• Available for immediate start • Willing to relocate • References available on request